```
> restart;#TheSun&Earth
```

```
> c := 299792458; LY := 365.25·24·3600·c; pc := 3.08567758·10^16; Rp := 6.371·10^6; Rso := 696340000;
  R := 3.667·10^22; Rs := 7.463764553·10^9·LY; 2·Rs; 13.79 − 4.6; 2.371797864·10^-4·LY; 2.418978218
  ·10^-4·LY; 2·7.463764553; RsoLY := Rso/LY;
```

$$c := 299792458$$

$$LY := 9.460730473 \times 10^{15}$$

$$pc := 3.085677580 \times 10^{16}$$

$$Rp := 6.371000000 \times 10^{6}$$

$$Rso := 696340000$$

$$R := 3.667000000 \times 10^{22}$$

$$Rs := 7.061266475 \times 10^{25}$$

$$1.412253295 \times 10^{26}$$

$$9.19$$

$$2.243894033 \times 10^{12}$$

$$2.288530094 \times 10^{12}$$

$$14.92752911$$

$$RsoLY := 7.360319607 \times 10^{-8} \quad (1)$$

```
> (Rp/pc)·120·10000;
  #Earth Radius growth  per year [pc = parsec] according to "Gravity" by Ioan N. Popescu (12.2) page
  576
```

$$0.0002477640583 \quad (2)$$

```
> int((4(π + 3/8)·(7.463764553)^2/(7π) r^-2 − 1/4 7.463764553/π r^-1 + (-4π + 2)/(28π), r = 0.362..13.79);
```

$$92.07421935 \quad (3)$$

```
> Λ := int((4(π + 3/8)·(7.463764553)^2/(7π) r^-2 − 1/4 7.463764553/π r^-1 + (-4π + 2)/(28π), r);
  # the increase due to the expansion of the space
```

$$\Lambda := -0.5939475112 \ln(r) - 0.1201207224\, r - \frac{35.63280450}{r} \quad (4)$$

```
> evalf(subs(r = 0.362, Λ)); evalf(subs(r = 13.79, Λ)); evalf(subs(r = 14.928, Λ)); evalf(subs(r = 9.19,
  Λ)); TΛ := evalf((subs(r = 13.79, Λ) - subs(r = 0.362, Λ))); evalf(-subs(r = 9.19, Λ) + subs(r
  = 13.79, Λ))
```

$$-97.87312865$$

$$-5.798909304$$

$$-5.785721803$$

$$-6.298699309$$

$$T\Lambda := 92.07421934$$

$$0.499790005 \quad (5)$$

```
> evalf((-subs(r = 9.19, Λ) + subs(r = 9.19000001, Λ))/10^-8);
  #the space growth rate in a period of  10 years 4.6 mild ago
```

$$0.2 \quad (6)$$

> $evalf\left(\frac{(-subs(r=13.79,\Lambda)+subs(r=13.7900001,\Lambda))}{10^{-7}}\right);$

#the space growth rate in a period of 100 years now days

0.01 **(7)**

> $evalf\left(\frac{(-subs(r=9.19,\Lambda)+subs(r=13.79,\Lambda))\cdot 696340000}{4.6\cdot 1000}\right)$;#the increase due **to** the spacethe expansion , of the Sun radius **in** a period of 4.6 billion years til now **in** Km

75657.3417 **(8)**

> $evalf\left((-subs(r=9.19,\Lambda)+subs(r=9.19000001,\Lambda))\cdot\frac{696340000}{10}\right)$; #the increase due **to** the expansion of the space of the Sun radius **in** a period of 10 years average per year 4.6 billion years ago $\frac{60\ cm}{year}$

0.06 **(9)**

> $evalf\left(\frac{(-subs(r=9.29,\Lambda)+subs(r=9.29000001,\Lambda))}{10}\cdot 6.371\cdot 10^{6}\right)$;#the increase due **to** the expansion of the space of the Earth radius **in** a period of 10 years average per year, 4.6 billion years ago $\frac{1.5\ mm}{year}$

0.0015 **(10)**

> $evalf\left(\frac{(-subs(r=13.79,\Lambda)+subs(r=13.79000001,\Lambda))}{10}\cdot 6.371\cdot 10^{6}\right)$;#the increase due **to** the expansion of the space of the Earth radius **in** a period of 10 years average per year nowadays $\frac{0.2\ mm}{year}$

0.0002 **(11)**

> $evalf\left(\frac{(-subs(r=13.79,\Lambda)+subs(r=13.7900001,\Lambda))}{100}\cdot 696340000\right);$

#tthe increase due to the spacethe expansion of the Sun in a period of 100 years average per year, nowadays $\frac{10\ cm}{year}$

0.01 **(12)**

> $evalf((-subs(r=9.19,\Lambda)+subs(r=13.79,\Lambda)));$

#Thus in the last 4.6 billion years the space growth was 499.79 milion LY

0.499790005 **(13)**

> $evalf\left(\frac{0.499790005}{4.6\cdot 1000}\cdot 696340000\right)$;#the space increase in Km of the Sun radius in 4.6 billion years time

75657.34177 **(14)**

> $evalf\left(\frac{(-subs(r=9.29,\Lambda)+subs(r=13.79,\Lambda))}{4.5\cdot 1000}\cdot 6.371\cdot 10^{6}\right);$

#the space increase in Km of the of the Earth Radius in 4.5 billion years

674.608738 **(15)**

> $\frac{75657.34177\cdot 1000}{Rso}$;# percentage growth of the Sun radius in 4.6 billion years

0.1086500011 **(16)**

> $\frac{674.608738\cdot 1000}{Rp}$;# percentage growth of the Earth radius in 4.5 billion years

0.1058874177 **(17)**

> $\frac{75657341.77}{4.6 \cdot 10^9}$; # *average growth per year of the Sun radius in period of 4.6 billion years ~16.45cm*

$$0.01644724821 \quad \textbf{(18)}$$

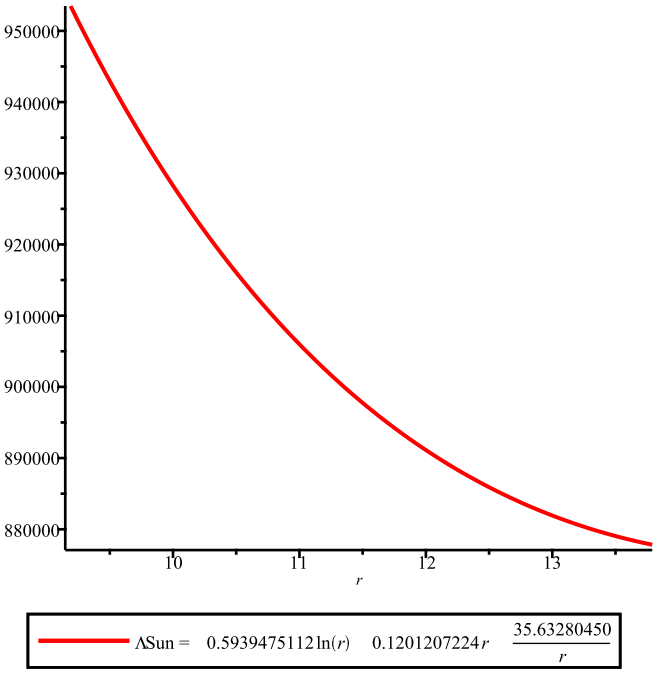
> $\frac{674.608738 \cdot 1000}{4.5 \cdot 10^9}$;# *average growth per year of the Earth radius in period of 4.5 billion years ~0.15mm*

$$0.0001499130529 \quad \textbf{(19)}$$

> $\Lambda Sun := \Lambda;$

$$\Lambda Sun := -0.5939475112 \ln(r) - 0.1201207224\, r - \frac{35.63280450}{r} \quad \textbf{(20)}$$

> $plot\left(-\frac{\Lambda Sun \cdot 696340}{4.6}, r = 9.19..13.79, color = red, legend = \left[typeset\left("\Lambda Sun = ", -0.5939475112 \ln(r) - 0.1201207224\, r - \frac{35.63280450}{r}\right)\right]\right);$



> # *We see that increase of the Sun radius due to the space growth from 4.6 billion years ago to present time, we see that the relative growth decrease;*

>